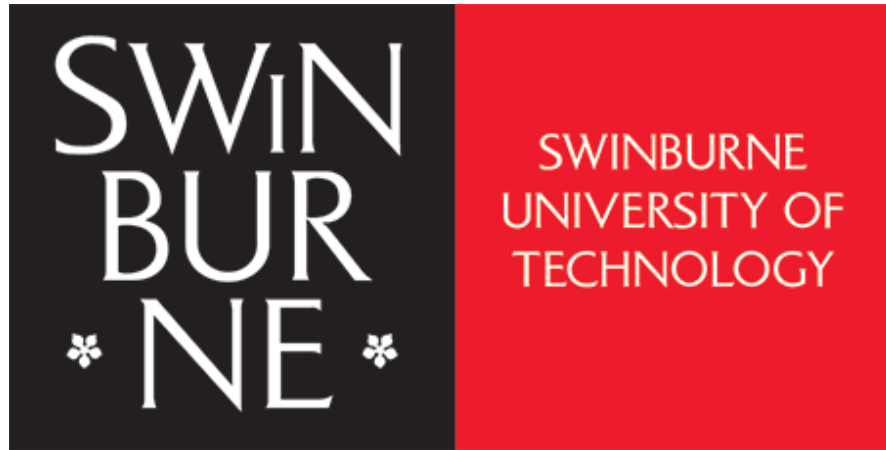




2026-HS1-COS10005-Web Development

Workshop Guide

Week 2 Workshop

Mercury Setup + HTML5 Markup & Validation

Objectives:

By the end of this workshop, students will be able to:

1. Write and structure basic HTML5 web pages.
2. Edit and format HTML content.
3. Set up and log in to the Mercury web server.
4. Upload HTML files to the Mercury server.
5. Test and view your web page via a browser and other tools

Exercises (10 marks):

Part 1: HTML Markup and Validation (45 mins) (3 marks)

Task 1: Create a HTML File

Step 1: Create a new text file and type the following HTML5 code.

```
<!DOCTYPE html>
<html Lang="en">
<head>
    <meta charset="utf-8" />
    <meta name="description" content="Web development" />
    <meta name="keywords" content="HTML Markup" />
    <meta name="author" content="put your name here" />
<title>My First HTML5 webpage</title>
</head>
<body>
    <!-- Web Development Unit Outline -->
    <!-- ALL HTML contents to be placed here -->
</body>
</html>
```

Note: This file can be used as a template for creating HTML5 web pages. All HTML contents should be placed between the <body> and the </body> tags.

Step 2: Save the file as **unitoutline.html**.

Task 2: Insert Content into Your HTML File

Step 1: Use VS Code, open the text file **unitoutline.txt**, which is available on Canvas.

Step 2: Copy everything in unitoutline.txt and paste it into unitoutline.html between the <body> tag and the </body> tag.

Step 3: Open unitoutline.html in a browser and see what it looks like.

Task 3: Structure Content with Semantic HTML Elements

Step 1: Play around with the code and Mark up the text contents in unitoutline.html using the HTML elements discussed in Lecture 2, e.g.,

- Headings <h#>...</h#>, Paragraphs <p>..</p>, Horizontal Rules <hr />
- Lists .. or ..., Tables <table>...</table>, Images and Anchors <a ... >...

For instance, headings can be marked up using heading elements:

```
<h1>HIT1091</h1>
<h2>Web DeveLopment</h2>
```

***Note:** Please use unitoutline.pdf, which is available on Canvas, as a guide to determine which HTML elements to use. The Swinburne logo is available on Canvas as unitoutline_logo.png.*

Part 2: Mercury Server Setup (25 mins) (3 marks)

Task 1: Set Up and Test Your Account on Mercury

Mercury is running the SSH service for both Terminal and File Transfer on port 22.

For Windows Users:

Step 1: Log onto the Mercury web server and set up your account.

Use 'PuTTY', an SSH (Secure Shell) client to log in to the Mercury web server.

***Note:** Available on most lab computers under Start > Programs > Network, or downloadable from <https://www.chiark.greenend.org.uk/~sgtatham/putty/latest.html>.*

Its host name is **mercury.swin.edu.au**, as presented in Figure 1.

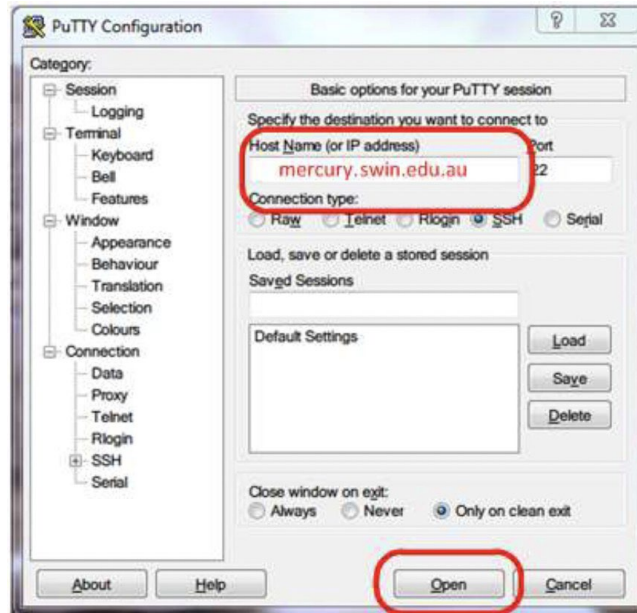


Figure 1. Accessing the Mercury Web Server on Windows

If the dialog presented in Figure 2 pops up, click on Yes.



Figure 2. PuTTY Security Alert

Step 2: Now enter your Mercury username: s<your Swinburne ID>, (for example, s1234567) and your SIMS password, as presented in Figure 3. In PuTTY, the password you type would not be visible.

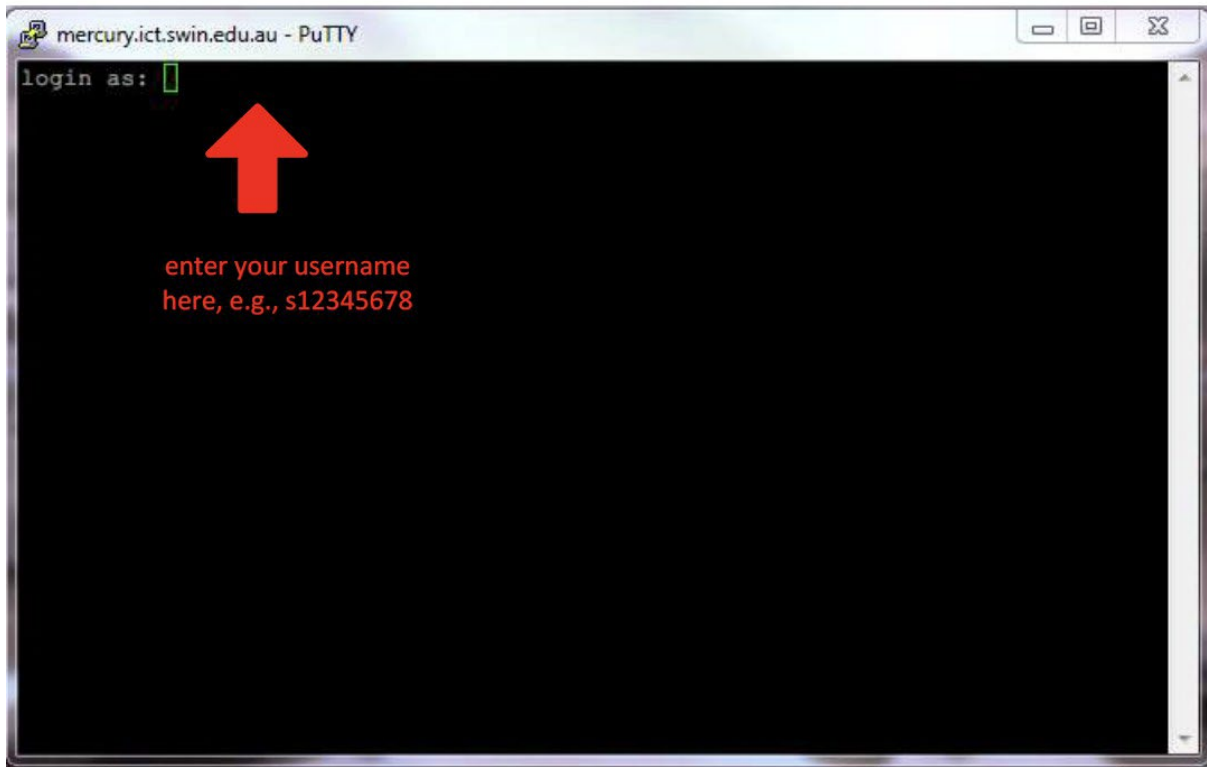


Figure 3. Entering Username

For Mac Users:

Step 1: Open the Terminal and use the built-in SSH client. Refer to Figure 4 for an example of the Terminal interface.

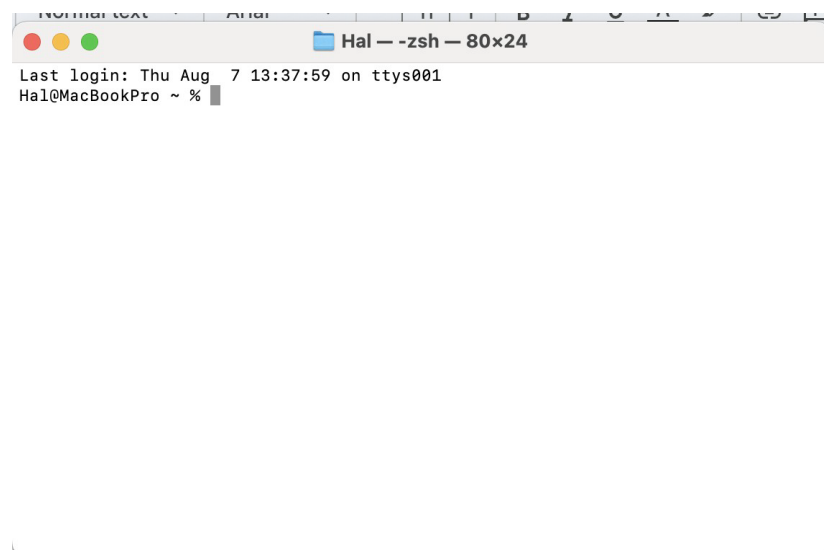


Figure 4. Mac Terminal Interface

Step 2: Connect to the server by using SSH

Now enter your Mercury username: s<your Swinburne ID>, (for example, s1234567, don't include the brackets), then click “**enter**” key. Be sure to include a lowercase 's' in front of your student number, as shown in Figure 5.

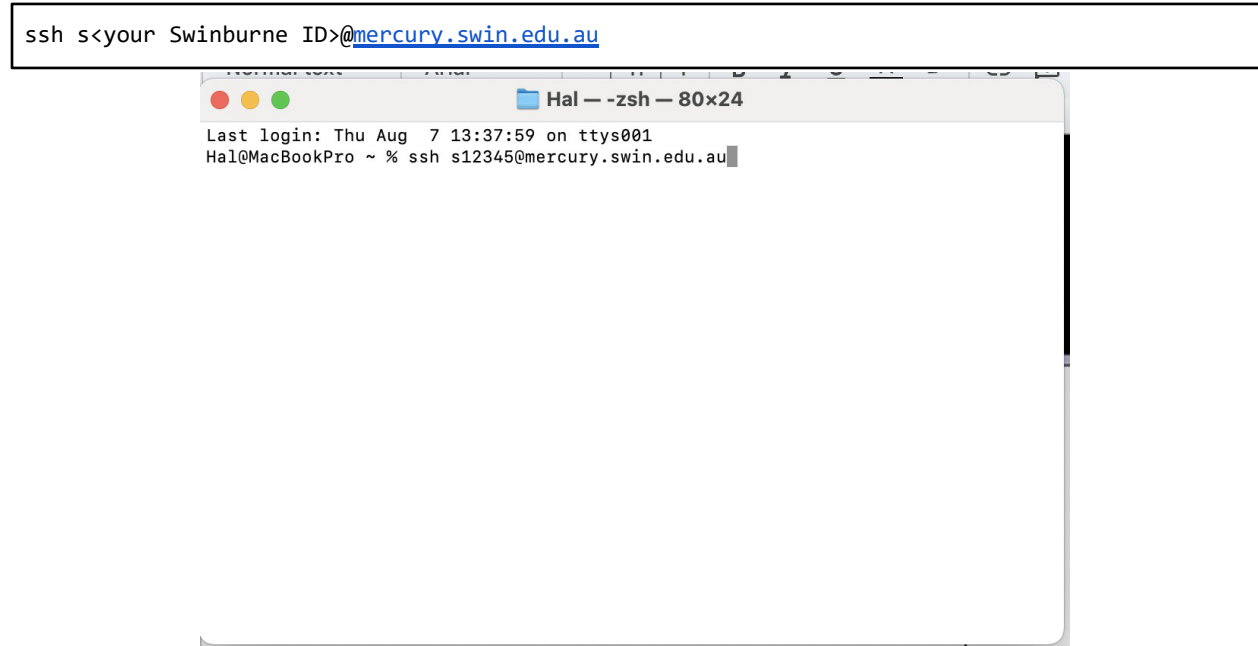


Figure 5. Enter your Mercury username

Then, it will ask you to input your password, the password you type would not be visible.

Part 3: Transferring Files to Mercury Server (25 mins) (3 marks)

To view and test a web page, you need to place it on Mercury properly. For this purpose we need to use **WinSCP (For Windows)** or or **Cyberduck (for Mac)**.

Note: WinSCP and Cyberduck are free for students to download.

WinSCP: <https://winscp.net/eng/download.php>

Cyberduck: <https://cyberduck.io/download/>

Task 1: Upload Files to Mercury

For Windows Users:

Step 1: Log in to the Mercury server using the WinSCP client

Use 'WinSCP', an SCP (Secure Copy Protocol) and SFTP client, to transfer files to the Mercury server, (available on most lab PCs under Start>Programs>Network), to log in to Mercury, with the host name “mercury.swin.edu.au”, user name s<your Swinburne ID> and your new Mercury password.

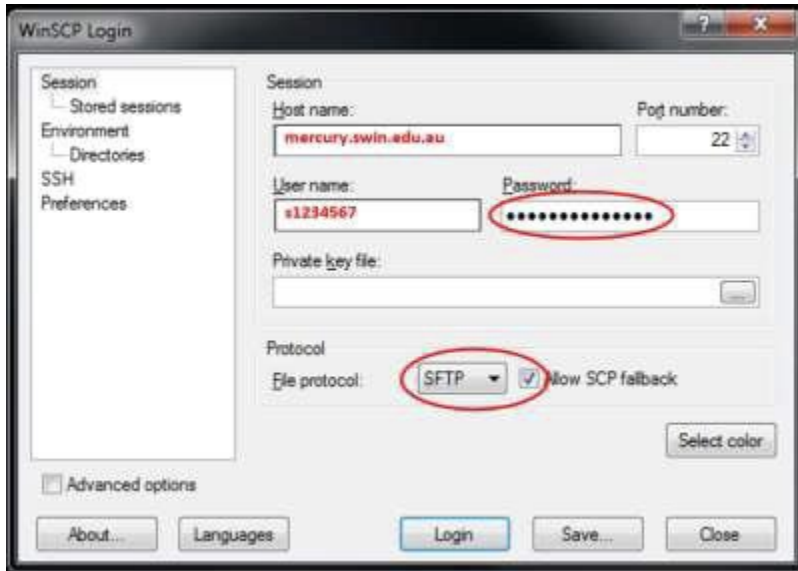


Figure 6. WinSCP client setup

After a successful log in, you will see the folder system of your local computer on the left the folder system of the Mercury server on the right. On the right, you will be able to see the unit folder “cos10005”.

Step 2: Create a folder (directory) to contain your web page files.

It is strongly recommended that you create a folder for each lab and assignment. Use WinSCP to access Mercury, create a folder/directory named “lab01” under ~/<your unit code>/www/htdocs folder on Mercury.

***Note:** Mercury web servers are case sensitive. It is recommended that when naming files or folders you always use lower case and avoid non-alphanumeric characters, such as a space.*

Step 3: Transfer your web pages to the web server.

To view and test your web pages, you must place the web page files into ~/cos10005/www/htdocs folder on mercury or any of its subfolders. Using WinSCP, drag and drop (or copy and paste) your file ‘myhtml1.html’ from your local computer to the ~/<your unit code>/www/htdocs/lab01 folder on the server.

For Mac Users:

Step 1: Log in to the Mercury server using the Cyberduck client

Use ‘Cyberduck’, an SFTP (SSH File Transfer Protocol) client, to transfer files to the server. To log in to Mercury, with the host name “mercury.swin.edu.au”, user name s<your Swinburne ID> and your password.

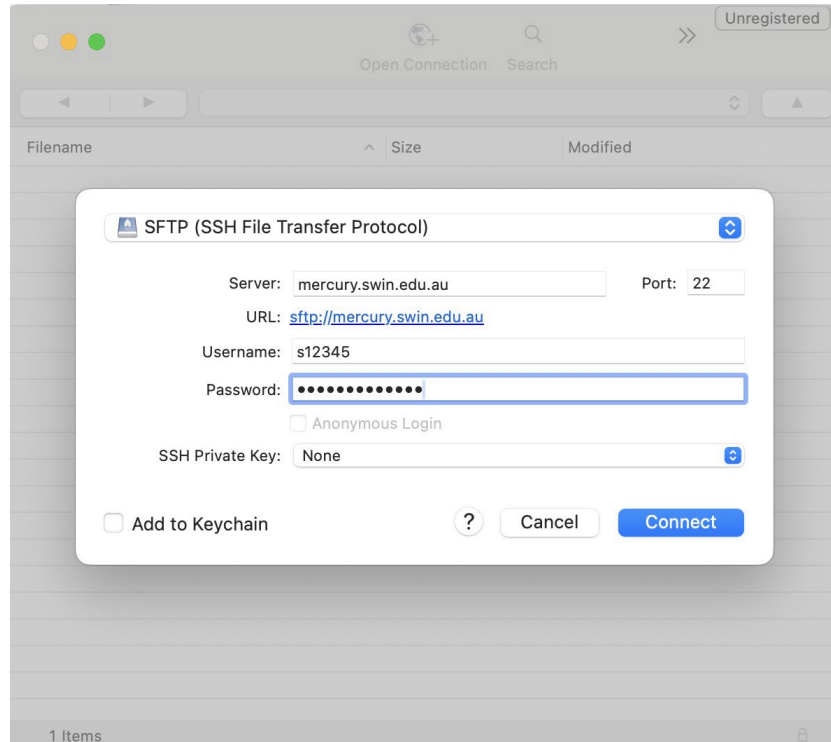


Figure 7. Cyberduck client setup

Step 2: Create a folder (directory) to contain your web page files.

It is strongly recommended that you create a folder for each lab and assignment. Use Cyberduck to access Mercury, create a folder/directory named “lab01” under `~/<your unit code>/www/htdocs` folder on Mercury.

Note: Mercury web servers are case sensitive. It is recommended that when naming files or folders you always use lower case and avoid non-alphanumeric characters, such as a space.

Step 3: Transfer your web pages to the web server.

To view and test your web pages, you must place the web page files into `~/cos10005/www/htdocs` folder on mercury or any of its subfolders. Using Cyberduck, drag and drop (or copy and paste) your file ‘myhtml.html’ from your local computer to the `~/<your unit code>/www/htdocs/lab01` folder on the server.

Part 4: Testing and Validating Web Pages (10 mins) (1 mark)

Task 1: View Your Web Page

To view your web page, use any web browser and type in the following address, `http://mercury.swin.edu.au/<your unit code>/s<your Swinburne ID>/<folder>/<filename>`

Please refer to the following examples to identify the URLs of your web pages.

Folder on Mercury Web Server	URL
~/cos10005/www/htdocs	http://mercury.swin.edu.au/cos10005/s1234567
~/cos10005/www/htdocs/index.html	http://mercury.swin.edu.au/cos10005/s1234567/index.html
~/cos10005/www/htdocs/lab01/myhtml.html	http://mercury.swin.edu.au/cos10005/s1234567/lab01/myhtml.html

Note: You can copy the URLs in the table, but remember to replace the unit codes and student id in the above examples with yours to obtain the URLs of your web pages on Mercury.

Note: When the browser authorization request dialog pops up, use your SIMS username and password to confirm access, NOT your mercury username and password.

Task 2: Test and Validate

You can use the validator at <http://validator.w3.org> to validate your unitoutline.html to Validate the page and fix any errors displayed and revalidate.

Reflection and Closing (10 minutes)

Allow students to share what they learned, found interesting, or had difficulty understanding.